YEAR 1 · TERM 1 · MODULE 3 · WEEK 1

Database management systems

Organisations run on data. This week you see why scattered files failed, what a database and a DBMS are, who works with them, and how the relational model organises data in tables linked by keys. You will protect data with constraints and transactions, and write your first SQL queries in a real database inside your browser.

Learning objectives

- Explain the problems of file-based systems and how a DBMS solves them.
- Distinguish data, information, database, DBMS, and database system.
- Identify the roles that work with a database.
- Read and design tables with primary keys, foreign keys, and N:M junction tables.
- Explain integrity constraints, transactions, and the three-schema architecture.
- Write SELECT queries with WHERE, ORDER BY, and JOIN.

Agenda

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From files to databases

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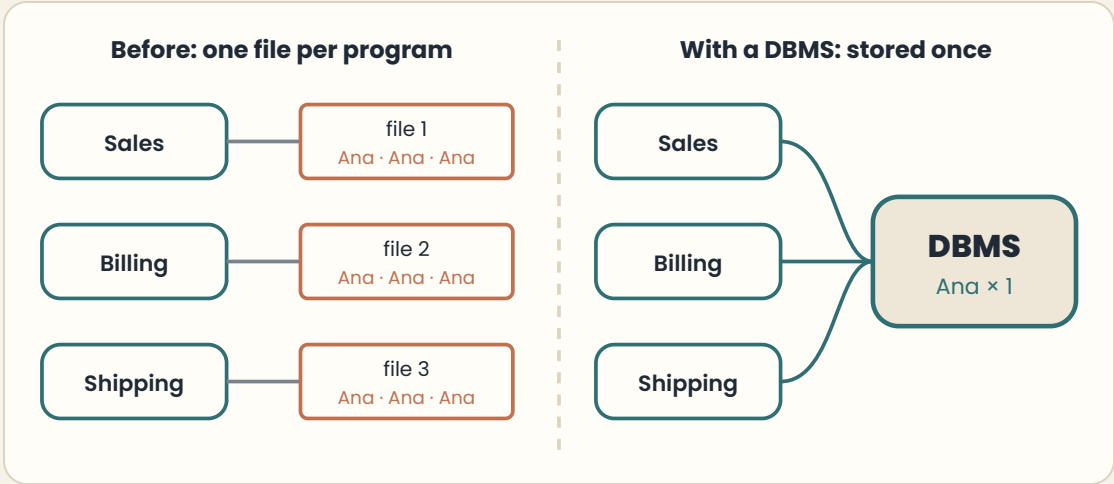
First SQL queries

LESSON 1

From files to databases

Key takeaways

- File-based systems duplicated data, so copies became inconsistent.
- Programs depended on file layouts: a layout change meant rewriting code.
- A DBMS stores data once, describes it, protects it, and shares it.



Think of it like this Picture every office in a company keeping its own paper address book. When a customer moves, someone has to update every book, and some will be missed. A database is one shared, well-organised address book that everyone consults.

LESSON 2

Data, information, and what a database is

Key takeaways

- Data + context = information; people turn information into knowledge.
- A database models a mini-world, is coherent, and has a purpose.
- The catalogue stores metadata: the structure and rules of the data.

Think of it like this A single puzzle piece is data. Seeing that it is part of the sky is information. Knowing where to start the puzzle next time is knowledge. The box with the picture on the lid is the metadata.

LESSON 3

What a DBMS does

Key takeaways

- A DBMS defines, constructs, manipulates, and shares databases.
- Database system = DBMS + data + application programs.
- Key advantages: less redundancy, access control, fast queries, recovery, integrity.

> **Think of it like this** A DBMS is like a librarian. You do not walk into the storage room yourself: you ask, and the librarian finds the book quickly, checks you may borrow it, records the loan, and makes sure two people do not take the last copy.

LESSON 4

Who works with a database

Key takeaways

- DBA: access, performance, security. Designer: structure, rules, views.
- Analysts specify; application programmers build and maintain.
- End users can be casual, parametric, sophisticated, or stand-alone.

Think of it like this Think of a theatre: the designer draws the stage, the administrator runs the building and holds the keys, programmers build the props, and the audience, the end users, are why the show exists.

LESSON 5

The relational model: tables, rows, and columns

Key takeaways

- Relation = table, attribute = column, tuple = row.
- Schema = structure; data (instance) = content at a moment.
- Other models exist: hierarchical, network, object-oriented, NoSQL.

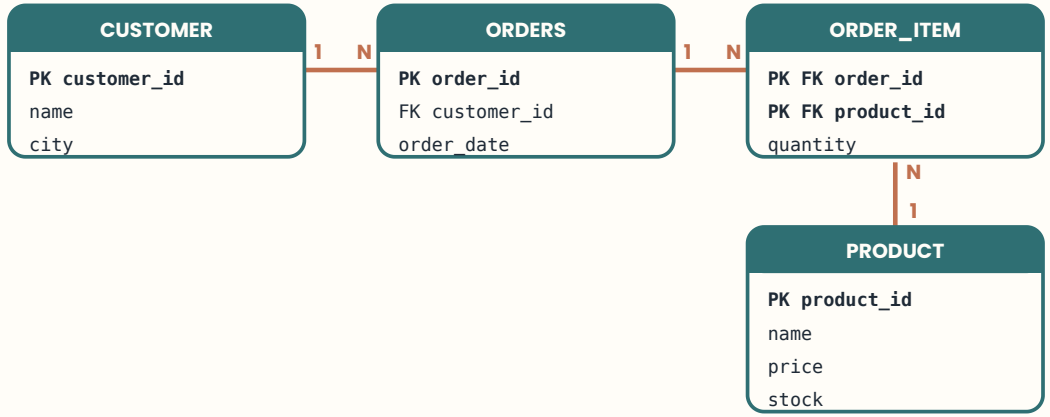
Think of it like this A table is like a well-designed form printed many times: the questions on the form are the columns (the schema), and each filled-in copy is a row (the data).

LESSON 6

Keys and relationships

Key takeaways

- PK: unique and not null; it identifies each row.
- FK: stores another table's PK; it creates the relationship.
- N:M relationships need a junction table with two FKs.



Think of it like this A primary key is like a national ID number: one per person, never shared. A foreign key is that same number written on a form somewhere else, so you can always find the person it refers to.

LESSON 7

Integrity, transactions, and safety

Key takeaways

- Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK.
- Transaction = all or nothing: COMMIT or ROLLBACK.
- Concurrency control, logs, backups, and least privilege keep data safe.

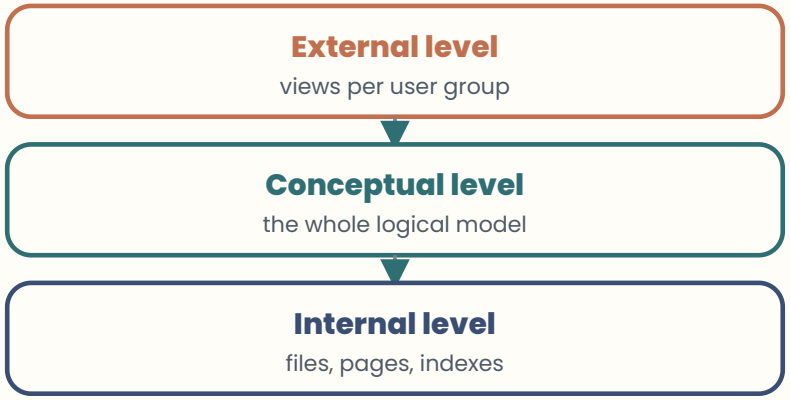
> **Think of it like this** A transaction is like a bank transfer: money leaves one account and arrives in the other, or nothing happens. You never want the money to leave and not arrive.

LESSON 8

Three levels, views, and indexes

Key takeaways

- External = user views; conceptual = logical model; internal = physical storage.
- Data independence lets one level change without breaking the others.
- Indexes speed up searches but cost space and update time.



Data independence: one level changes without breaking the others.

Think of it like this A restaurant: diners see the menu (external), the chef works from recipes (conceptual), and the pantry is organised in shelves and fridges (internal). Reorganising the pantry should not change the menu.

LESSON 9

First SQL queries

Key takeaways

- SELECT columns FROM table WHERE condition.
- ORDER BY sorts; JOIN … ON combines tables through keys.
- UPDATE and DELETE always need a WHERE clause.

Think of it like this SELECT is like asking a librarian: "show me the titles and authors (SELECT) from the history shelf (FROM) published after 2000 (WHERE)".

Checkpoint

1. What is the main risk of storing the same customer in several files?

A. The copies become inconsistent · B. The files load faster · C. Programs need less code

2. Which statement is information, not just data?

A. "Customer 2 lives in Lima" · B. "Lima" · C. "2"

3. A data analyst writes her own SQL queries and small tools to study sales. What kind of end user is she?

A. Sophisticated · B. Parametric · C. Casual

Keep going

Review your flashcards when they are due, use the subject's tools, and retake the quiz until you score 80% or more.